Scalable Probit Calibration

Peter Cotton*

August 2026 — working draft, revised

Abstract

Probit models are a cornerstone of economic, choice, and contest theory, and machine learning runs on the same latent-utility family. Yet the many-alternative form sees little use beyond small menus: its probabilities have no closed form, the standard simulator prices one alternative at a time, and calibration, meaning the recovery of the utilities at which model shares match observed shares with the factor covariance given and fixed, has been out of reach at scale. We give an algorithm that calibrates a two-factor probit model with a thousand alternatives in under a minute on a laptop. Its cost grows linearly from there, reaching ten thousand alternatives in eleven minutes, where extrapolation from our plain per-alternative simulator implementation puts matched-accuracy simulation-based calibration on a weeks scale. Conditionally on the factors the alternatives are independent, so a single shared survival product yields every choice probability in one differentiable pass, and the inverse problem this solves is well posed, with shares determining utilities up to a common shift. Maximum observed share discrepancies stay below one part in a thousand from five to five thousand alternatives, and the experiments illustrate where the factor approximation becomes limiting.

1 Introduction

Probit models are a cornerstone of economic, choice, and contest theory: Thurstone's comparative judgment, the ordered and binary probits of applied microeconometrics, and the Lazear–Rosen tournament are all the same Gaussian latent-performance model. Machine learning runs on the same family: a softmax output layer is its Luce–Gumbel member, and Thurstone–Mosteller pairwise comparison, the probit member, underlies rating systems and preference learning.

Such models are calibrated to observed shares so that the recovered utilities can answer questions the data do not: elasticities, welfare, pricing, the effect of adding or removing an alternative. The models used for this, essentially universally, are the logit family: multinomial logit, nested logit, and above all random-coefficients logit in the BLP tradition [11, 4, 12]. They are universal for one reason. They can be computed.

Choice modeling did not start there: it started with Thurstone's 1927 model [1] of jointly Gaussian utilities with unrestricted correlation, known in econometrics as multinomial probit (MNP).

Probit probabilities are $(N-1)$-dimensional Gaussian integrals with no closed form. The standard evaluator is the Geweke–Hajivassiliou–Keane (GHK) simulator [2, 3, 4]: it writes the

*Every number in this paper is produced by a committed, seeded script; see the Reproducibility section and `https://github.com/microprediction/kinetics`.

probability that alternative i beats the rest as an orthant probability of the difference utilities, and estimates it by sampling truncated normals sequentially along a Cholesky factor. That is a separate simulation per alternative, with $R^{-1/2}$ error in the number of draws, and it is why probit survives mainly on small menus: the field kept the computable models and gave up the model it started from.

The concession is not free. Logit-family substitution is structurally restricted, and the calibration apparatus that demand estimation is built on exists on the logit side only. One measure of the cost, from one deliberately misspecified synthetic design with oracle factor loadings (Section 4): in the largest-deletion stratum, plain logit misallocates about 23% of the demand released by a deleted alternative, against 5% for factor probit, the closest of the four candidate structures.

This paper's point is that the barrier is not real for the covariance structure applied work most often actually has, the factor model $\Sigma = VV^\top + \mathrm{diag}(D)$. The goal is **calibration**: given observed shares and a fixed, known factor structure (V, D), recover the utilities at which the probit model reproduces them. This is the probit analogue of the inversions demand estimation runs on the logit side every day [11, 12]; estimating (V, D) is the outer problem, discussed in Section 6. We solve it at $N = 1000$ in under a minute, and the inverse problem is globally well-posed (Theorem 1: shares determine utilities uniquely up to location). Three ingredients make it work:

1. **A fast forward map** (utilities $\rightarrow$ shares): all N probabilities at once in $O(QNL)$ for Q factor nodes and L lattice points. All 5000 shares of an $N = 5000$ problem take 21 seconds, where our plain pseudorandom per-alternative GHK baseline extrapolates (in N and in draws) to roughly fifty hours at matched accuracy. Calibration needs many forward evaluations; this is what makes them affordable.

2. **Resampling-free derivatives**: Jacobian–vector products in $O(QNL)$; in max-wins coordinates the Jacobian is a weighted graph Laplacian, so reduced systems are symmetric positive definite. This supports implicit differentiation and Newton–Krylov solvers; the calibration reported below uses a cheaper damped Jacobi iteration.

3. **A single shared construction**: conditionally on the k factors the alternatives are independent, and a lattice survival-product identity computes every alternative's integral from one *shared survival field*, built once and reused for shares, slopes, and all N single-removal counterfactuals.

Prior art. The nearest methods, and what each lacks:

approach	all N shares	calibration
GHK simulation [2, 3]	N separate runs, $R^{-1/2}$ error	not attempted at scale
direct utility simulation	whole vector per draw, $R^{-1/2}$	no derivatives, no inversion
analytic approximations [7, 8]	per-alternative	no
factor quadrature [5, 6]	still per-alternative	no
Bayesian factor MNP [18]	simulation inside estimation	posterior, not a share map
neural amortization [14]	fast, uncontrolled error	no
BLP-side inversion [11, 12]	—	logit family only
this paper	one $O(QNL)$ pass	Thm. 1; <1 min at N=1000

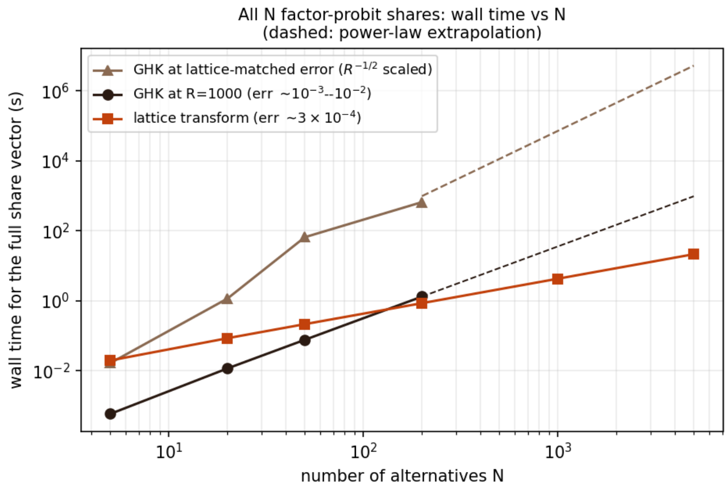

Figure 1: Wall time for the full N-alternative share vector as N grows ($k = 2$ factors). The upper GHK line rescales R by the squared error ratio ($R^{-1/2}$ scaling) so both methods deliver the same accuracy; at $N = 5000$ it extrapolates to roughly fifty hours against the lattice's 21 seconds (Section 2). Dashed segments are power-law extrapolations; all other points are measured (Table 1).

Figure 1 shows the running-time consequence as the number of alternatives grows. The per-alternative simulator's cost crosses the lattice transform near $N \approx 200$, where its maximum-coordinate error is about 23 times larger (Table 1); matching the lattice error would take roughly 500 times as many draws under $R^{-1/2}$ scaling. By $N = 5000$ the lattice computes all shares in 21 seconds; the extrapolated matched-accuracy comparison is in Section 2.

Scope. The method needs the factor form: where Σ has no adequate low-rank-plus-diagonal approximation in the choice-relevant quotient space, simulation at exact Σ remains the tool (Section 4). At accuracy targets near 10^{-3}, plain direct utility simulation is a genuinely competitive forward baseline (Section 2), and the decisive advantages lie elsewhere: error decay below that level, reproducibility, resampling-free derivatives, and the calibration machinery.

The one-pass identity comes from an unlikely literature: inferring ability from win probabilities in multi-entrant races [10]. That is perhaps why it has not been assembled for probit before, despite each ingredient being classical.

Model. Throughout,

$$U_i \;=\; \mu_i + v_i^\top f + \sqrt{D_i}\,\varepsilon_i, \qquad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0,1) \text{ i.i.d.}, \quad D_i > 0, \tag{1}$$

i.e. $\Sigma = VV^\top + \operatorname{diag}(D)$; the chosen alternative is the argmax of U. Throughout, (V, D) is supplied and held fixed; calibration solves for μ alone. Factor dimension reduction for probit is classical: one-factor quadrature goes back to Butler and Moffitt [5], and factor-analytic probit to Elrod and Keane [6]. The contribution is the all-alternatives assembly, the inverse transform, the derivative structure, and evidence for all of it.

Proposition 1 (Conditional identity)**.** *Under* (1)*, with $g_i(\cdot \mid f)$ and $F_i(\cdot \mid f)$ the conditional density and CDF of U_i given f,*

$$p_i \;=\; \mathbb{E}_f\left[\,\int g_i(x \mid f)\prod_{j\neq i} F_j(x \mid f)\,dx\right], \qquad \sum_i p_i = 1, \tag{2}$$

and the product over $j \neq i$, for all i simultaneously, is obtained from $\log F_{\text{field}} = \sum_j \log F_j$ by subtraction.

Proof. Conditional on f the U_j are independent, so $\mathbb{P}(U_i \text{ maximal} \mid f) = \int g_i \prod_{j\neq i} F_j\,dx$; average over f. The events partition (ties are null for continuous laws), giving $\sum_i p_i = 1$. ☐

Proposition 2 (Complexity)**.** *With Q factor nodes and L lattice points, the forward share vector costs $O(QNL)$; the full single-removal ensemble $\{P(j \text{ chosen} \mid i \text{ removed})\}_{i,j}$ costs $O(QN^2L)$ arithmetic and $O(N^2)$ storage, reusing one conditional field construction.*

Proof. Per node: the field log-product costs $O(NL)$, each alternative's integral $O(L)$ after subtraction, giving $O(NL)$; the removal ensemble evaluates N survivor integrals for each of N removals, $O(N^2L)$, over the same cached field. ☐

Proposition 3 (Identification and differentiability)**.** *$p(\mu + c\mathbf{1}) = p(\mu)$, so $J = \partial p/\partial\mu$ has $\mathbf{1}$ in its null space and μ is identified only up to location. With B an $N \times (N-1)$ basis of $\mathbf{1}^\perp$ and $\mu = B\eta$, share inversion and implicit differentiation are taken in reduced coordinates: $d\eta/d\theta = -(B^\top JB)^{-1}B^\top\partial p/\partial\theta$, provided $B^\top JB$ is nonsingular.*

Proof. Invariance holds because a common location shift does not change the argmax; differentiating $p(\mu + c\mathbf{1}) = p(\mu)$ in c gives $J\mathbf{1} = 0$. Differentiating $p(B\eta(\theta), \theta) = \hat{p}$ in θ and projecting with $B^\top$ gives the reduced formula under the stated regularity. ☐

Theorem 1 (Shares determine utilities)**.** *For model* (1) *with $D_i > 0$ and fixed known (V, D), in max-wins coordinates, define for $i \neq j$*

$$w_{ij} \;=\; \mathbb{E}_f \int g_i(x \mid f)\, g_j(x \mid f) \prod_{\ell\neq i,j} F_\ell(x \mid f)\,dx \;>\; 0.$$

Then $J = \partial p/\partial\mu$ satisfies $J_{ij} = -w_{ij}$ and $J_{ii} = \sum_{j\neq i} w_{ij}$, i.e.

$$J \;=\; \sum_{i<j} w_{ij}(e_i - e_j)(e_i - e_j)^\top, \qquad h^\top Jh \;=\; \sum_{i<j} w_{ij}(h_i - h_j)^2,$$

a weighted graph Laplacian with everywhere-positive weights: J is symmetric, positive semidefinite, has $\mathbf{1}$ as its only null direction, and $B^\top JB \succ 0$ for every full-rank basis B of $\mathbf{1}^\perp$. In English: w_{ij} is the probability density of a photo-finish between i and j, a tie for the win with everyone else behind, and moving utility from j to i moves share along every photo-finish edge at exactly that rate. The alternatives form a photo-finish graph*: complete, with strictly positive weights, because Gaussian utilities give every pair some chance of a dead heat. The theorem's conclusions are then graph facts. Constants span the null space because the quadratic form $\sum_{i<j} w_{ij}(h_i - h_j)^2$ sees only differences: shares cannot identify the overall utility level, only gaps, which is economically obvious and here algebraically automatic. Positive weights make the graph connected, so the*

constant vector is the only *null direction, the gradient map is strictly monotone as an operator on $\mathbf{1}^\perp$ (not coordinatewise), and shares determine utilities uniquely. And symmetry plus positive definiteness make every reduced Newton system SPD, so conjugate gradients apply with the $O(QNL)$ Jacobian–vector product of Proposition 4 playing the role of a fast Laplacian multiply. Structurally, calibration is solving a diffusion on the photo-finish graph, and it is easy for the same reason resistor networks and heat equations are. None of this is specifically Gaussian: any conditionally independent additive random-utility race with everywhere-positive smooth densities has a weighted-Laplacian Jacobian and an interior-simplex diffeomorphism modulo translation; Gaussian factor probit is the computationally important specialization. Moreover $p = \nabla G$ for the social-surplus function $G(\mu) = \mathbb{E}\max_i\{\mu_i + \xi_i\}$ (the Williams–Daly–Zachary / McFadden identity [15]), and the restriction of p to $\mathbf{1}^\perp$ is a smooth bijection onto the interior of the probability simplex: every strictly positive target $\hat{p}$ has exactly one mean-zero utility vector.*

Proof. Differentiating (2) in μ_j and integrating by parts gives $J_{ij} = -w_{ij}$ for $i \neq j$; the diagonal follows from $J\mathbf{1} = 0$ (Proposition 3), and positivity of w_{ij} from positivity of Gaussian densities. The quadratic form is then immediate, and it vanishes only for constant h. For inversion, minimize $H(\mu) = G(\mu) - \hat{p}^\top \mu$ over $\mathbf{1}^\perp$: H is strictly convex there (its Hessian is the reduced J), and since the shocks have zero mean, $G(\mu) \geq \max_i \mu_i$, so

$$H(\mu) \;\geq\; \max_i \mu_i - \hat{p}^\top \mu \;=\; \textstyle\sum_i \hat{p}_i(\max_j \mu_j - \mu_i) \;\geq\; \hat{p}_{\min}\,(\max_i \mu_i - \min_i \mu_i) \;\geq\; \hat{p}_{\min}\,\|\mu\|_2/\sqrt{N}$$

on $\mathbf{1}^\perp$, where the last step uses that a zero-sum vector has $\max_i \mu_i \geq 0 \geq \min_i \mu_i$, so the range dominates $\|\mu\|_\infty \geq \|\mu\|_2/\sqrt{N}$; hence H is coercive. Since $D_i > 0$ the utilities have a smooth full-support joint density and ties are null, so differentiating under the expectation gives $\nabla G = p$. At the unique constrained minimizer, $B^\top(p(\mu) - \hat{p}) = 0$, so $p(\mu) - \hat{p} \in \operatorname{span}\{\mathbf{1}\}$; both vectors sum to one, so the coefficient is zero and $p(\mu) = \hat{p}$. Smoothness of the inverse follows from the inverse-function theorem, since $B^\top J B \succ 0$ everywhere. $\square$

Remark 1. Social-surplus convex duality for discrete choice is classical [16, 17]; what is new here is not well-posedness but its scalable realization. Each identity in the theorem is also checked numerically in the repository: the w_{ij} representation against finite differences to 6×10^{-9}, symmetry to 4×10^{-12}, and positive definiteness of the reduced numerical Jacobian.

Proposition 4 (Jacobian–vector products in $O(QNL)$)**.** *With hazards $\lambda_j = g_j/F_j$, $\Lambda = \sum_j \lambda_j$, and $A = \sum_j h_j\lambda_j$ (all conditional on f, all functions of x), the Jacobian of the exact max-wins map acts on a direction h as*

$$(Jh)_i \;=\; \mathbb{E}_f \int g_i\Big(\prod_{j\neq i} F_j\Big)\,(h_i\Lambda - A)\,dx,$$

where the field sums Λ and A are formed once per node and lattice point, so all N components cost $O(NL)$ per node. In English: a direction h perturbs alternative i's share only through the gap between its own hazard weight $h_i\Lambda$ and the field average A. Translation invariance is visible ($h = \mathbf{1}$ gives $A = \Lambda$, hence $Jh = 0$), and the weighted-Laplacian form of Theorem 1 follows by expanding A. This enables Krylov solves of the reduced system in Proposition 3 without forming J densely at $O(QN^2L)$.

Proof. Write $R_i = \prod_{j\neq i} F_j$. For the own derivative, $\partial_{\mu_i} g_i(x) = -g_i'(x)$ (a location shift), and integrating by parts, $\int(-g_i')R_i\,dx = \int g_i R_i'\,dx = \int g_i R_i \sum_{j\neq i}\lambda_j\,dx = \int g_i R_i(\Lambda - \lambda_i)\,dx$. For $j \neq i$, $\partial_{\mu_j} F_j = -g_j$, so the cross derivative is $-\int g_i g_j \prod_{\ell\neq i,j} F_\ell\,dx = -\int g_i R_i \lambda_j\,dx$. Summing against h, $(Jh)_i = \int g_i R_i[\,h_i(\Lambda - \lambda_i) - \sum_{j\neq i} h_j\lambda_j\,]dx = \int g_i R_i(h_i\Lambda - A)\,dx$; average over f. $\square$

The differentiated map. Three maps must be distinguished: the exact max-wins map $p^{\max}(\mu)$; the reflected min-wins map $p^{\min}(a)$ used by the implementation, related by $a = -\mu$ so that $D_\mu p^{\max}[h] = D_a p^{\min}[-h]$; and the implemented approximation $p_{Q,L} = \tilde{p}_{Q,L}/(\mathbf{1}^\top \tilde{p}_{Q,L})$, whose derivative carries a normalization correction: with $v = D\tilde{p}[h]$ and $s = \mathbf{1}^\top \tilde{p}$,

$$Dp_{Q,L}[h] \;=\; \frac{v - p_{Q,L}\,(\mathbf{1}^\top v)}{s}.$$

Since $\mathbf{1}^\top v = 0$ analytically, the correction is of the order of the quadrature defect. One further distinction, raised by a careful reading: integration by parts is a continuum identity, so the Laplacian formula of Proposition 4 is the exact derivative of the *continuum* map, not of the finite rectangle sum. The exact derivative of the frozen-grid sum replaces the own-term by $(x-m_i)/D_i$ (no integration by parts), costs the same $O(QNL)$, and ships in the repository (`form="grid"`). Measured at production resolution ($L = 1501$) the two forms agree to 1.7×10^{-14}; at $L = 51$ the continuum form is off by 2.6×10^{-5} while the grid form still matches finite differences. So the Laplacian form is an extremely accurate symmetric surrogate at the resolutions used, and exact differentiation of the returned map is available when needed. Finally, the implemented inversion rebuilds the lattice envelope from the current utilities at each iteration, holds it fixed while the forward map and diagonal slopes are evaluated, and projects the utilities to mean zero after every update; the slopes ignore the moving envelope and the output normalization and are therefore Jacobi quasi-Newton preconditioners, not the exact diagonal of the adaptive map's Jacobian.

Tail-stable implementation. Tail stability requires the log domain throughout: $\log\Phi$ evaluated directly (forming $1 - \Phi(z)$ by literal subtraction rounds to zero near $z \approx 8.3$ through cancellation), the analytic $\log g$, and hazards as $\exp(\log g - \log\Phi)$; the Gaussian inverse Mills ratio is unbounded but grows only linearly in the extreme tail, rather than exponentially, which is the numerically important property. Flooring a density before taking its logarithm while the exact log-survival stays finite produces spurious e^{+92} hazards on problems as mild as $N = 8$ with heterogeneous D; that configuration is a regression test. In the forward map the same discipline lets deep-tail shares resolve to genuine small positives rather than floored zeros.

Beyond Gaussian factors. The factors need not be Gaussian: (2) requires only conditional independence and an integrable factor law, so Student, mixture, or empirical factor distributions are admissible with appropriate nodes, and the base densities may be arbitrary and alternative-specific. One caution for asymmetric laws: the implementation works in the reflected (min-wins) race, and $-U$ has the same distribution as the reflected model only when the factor and base laws are symmetric; for asymmetric laws the reflected distributions must be used explicitly.

Numerical method. The benchmarks use $L = 1501$ lattice points on the adaptive interval $[\min_{i,q} m_i(f_q) - 8\max_i \sqrt{D_i},\ \max_{i,q} m_i(f_q) + 8\max_i \sqrt{D_i}]$, where $q = 1, \ldots, Q$ indexes the retained factor nodes.

Factor integration: the GHK benchmark (experiment 13) uses product Gauss–Hermite of order 15/11/9 for $k = 2/3/4$; the boundary experiment (experiment 14) uses order 15 per dimension for all $k \le 4$, which after pruning weights below 10^{-7} of the maximum leaves $Q = 145$, 1317, and 10929 nodes for $k = 2, 3, 4$, making the exponential growth of Q with rank explicit.

Scrambled-Sobol nodes (2^{13} points, fixed seed) are used for $k > 4$: fixed-node and reproducible (a fixed-seed randomized Sobol construction), rather than deterministic in the strict sense.

Survival floors sit at 10^{-300} and the quadrature output is normalized by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2); statements about derivatives refer to derivatives of this approximation, whose fidelity to exact MNP derivatives is governed by L and Q.

Algorithm. Orientation: the implementation computes the reflected min-wins race on $a = -\mu$. Per retained factor node q: form conditional locations $m_i(f_q) = a_i + v_i^\top f_q$; on the uniform lattice evaluate $\log S_i = \log \Phi(-(x - m_i)/\sqrt{D_i})$; accumulate the field sum $\log S_{\text{field}} = \sum_j \log S_j$; recover each alternative's integrand as $g_i \exp(\log S_{\text{field}} - \log S_i)$; integrate by the rectangle rule with Δx weights and no endpoint modification (differing from the trapezoidal rule only by negligible endpoint terms). Pruned factor weights are not renormalized explicitly; the final normalization conditions the quadrature on the retained node set. The two contributions to the pre-normalization defect separate cleanly: the pruned factor weight is $\delta_{\text{factor}} = 2.39 \times 10^{-8}$ at $k = 2$ (analytic), and the measured $|1 - \mathbf{1}^\top \tilde{p}| = 2.4 \times 10^{-8}$ equals it, so lattice truncation contributes below that level. Including the conditional-location build, the complete forward cost is $O(QN(k + L))$, which reduces to $O(QNL)$ for $k \ll L$. Inversion: Newton step $\mu \leftarrow \mu - \text{damp} \cdot r/\partial_i$, with r the log-share residual on well-resolved alternatives, ∂_i the analytic own-location log-slope, steps capped at $\sqrt{D_i + \|v_i\|^2}$, damping halved (floor 0.25) whenever the residual grows by more than 20%, tolerance 10^{-6} on the identified set, at most 50 iterations, followed by mean removal.

Inversion. Given target shares $\hat{p}$ and (V, D), the inverse transform finds mean-zero μ with $p(\mu) = \hat{p}$ by a damped Jacobi quasi-Newton iteration: all coordinates updated simultaneously with own-location slopes of the unnormalized lattice map as the preconditioner (available from the same pass), the field refrozen each iteration, a warm start from the independent-field inverse using total variances, and a tail-aware tolerance (alternatives with target shares below $10^{-4}/N$ are matched best-effort; by Theorem 1 they remain uniquely determined, but they are statistically weakly resolved, which is also a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear). Strictly zero observed shares lie outside the theorem: boundary targets correspond to utilities diverging to $-\infty$, so real assortment data requires pseudocounts or regularization, not merely the implementation floor. Convergence in a handful of iterations is an empirical observation on the tested problems, not a theorem. The design follows the original transform's inversion [10] and the calibration practice of [19].

2 Benchmark against GHK

Correctness anchors. On the binary case, where MNP has a closed form, the lattice transform agrees to 3.3×10^{-16}, and our GHK implementation (validated before being compared against) to 2×10^{-14} at $R = 2 \times 10^5$ (an easy anchor for both methods, useful for code validation rather than as evidence about the hard regime). At $N = 5$ both methods sit within the noise of a 10^7-draw Monte Carlo reference, and the shipped `thurstone.FactorRace` agrees with the benchmark kernel to 1.1×10^{-5}.

Error metric. Throughout, $\max_i |\hat{p}_i - p_i|$ against a simulation reference. A maximum over N coordinates tends to grow with N at fixed per-coordinate accuracy, and the reference's own maximum-coordinate noise behaves likewise; mean-absolute and total-variation summaries are committed alongside (experiment 16): at $N = 1000$ the lattice's mean absolute error is 5.6×10^{-6} and total variation 2.8×10^{-3} against a fresh 2×10^6-draw reference.

N	lattice transform	direct MC, matched time	GHK ($R = 1000$)
5	20 ms, err 5.7×10^{-4}	err 4.9×10^{-4}	1 ms, err 3.1×10^{-3}
20	86 ms, err 4.4×10^{-4}	err 1.4×10^{-3}	12 ms, err 4.4×10^{-3}
50	217 ms, err 2.7×10^{-4}	err 8.4×10^{-4}	77 ms, err 7.9×10^{-3}
200	860 ms, err 3.0×10^{-4}	err 6.2×10^{-4}	1323 ms, err 6.8×10^{-3}
1000	4.3 s, err 6.8×10^{-4} †	err 6.2×10^{-4}	—
5000	21.5 s, err 5.0×10^{-4} †	—	—

Table 1: Full share vector, $k = 2$ factors, one problem per size, generated as $\mu_i \sim N(0, s^2)$ with spread $s = 1$ through $N = 200$ and 1.5 above, $v_{ir} \sim N(0, 0.25/k)$, $D_i \sim U[0.5, 1.5]$. Sub-second timings are warmed medians of three runs; the direct-MC column runs the draw-argmax simulator at wall time matched to the lattice call (experiment 16). Reference: 2×10^6 MC draws through $N = 200$; 5×10^5 draws at $N = 1000$ and 5000 (†: there the reference's own maximum-coordinate noise is of comparable order, so these entries bound the lattice error only to that order). A replication study at common spread 1.0 (experiment 16; ten problems per size at $N \leq 200$, three at $N = 1000$, twin independent 2×10^6-draw references per problem) gives worst-case lattice error 3.5–4.6×10^{-4} at every size, at or below the references' own replicate-to-replicate noise. Lattice error stayed below 10^{-3} throughout; GHK cost crossed over by $N \approx 200$.

Matched-accuracy extrapolation. Fitting a power law to the reference implementation's timings over $N \in [20, 200]$ ($\alpha \approx 2.0$; a lower bound, the asymptotic per-share cost being cubic) and combining with $R^{-1/2}$ error scaling suggests that matching the lattice's 5×10^{-4} at $N = 5000$ would take this implementation on the order of fifty hours (0.27 h at $R = 1000$, times $(6.8 \times 10^{-3}/5 \times 10^{-4})^2 \approx 185$), against 21 seconds. The baseline is a straightforward single-threaded Python GHK: per-alternative Cholesky, plain pseudorandom uniforms, one realization per problem, shares normalized by their sum. It establishes the scaling of the per-alternative simulation approach, not a field-wide impossibility.

Direct simulation baseline. The strongest cheap baseline is *direct utility simulation*: draw utilities, take the argmax, average. It produces the whole share vector per draw and, unlike GHK, does not pay per alternative. We ran it at wall time matched to the lattice call (experiment 16): its maximum-coordinate error was 1.5–4× the lattice's at every tested N (6.2×10^{-4} versus 3.9×10^{-4} at $N = 1000$, 0.5M draws in the matched 4.1 seconds). At the 10^{-3} accuracy level, then, raw forward speed does not by itself separate the methods.

The separation is in error decay (experiment 17, $N = 200$, $k = 2$). Holding the factor rule fixed, lattice self-convergence is 2.7×10^{-8} at $L = 101$ and at machine precision from $L \approx 200$; the convergence observed over the tested range is near-exponential, and for these analytic, rapidly decaying integrands the lattice error becomes negligible very quickly, leaving the L-error subdominant. Holding L fixed, the Gauss–Hermite sweep decays from 5.3×10^{-4}

(order 3) to 5.9×10^{-9} (order 15). The pre-normalization defect $|1 - \mathbf{1}^\top \tilde{p}\,|$ is 2.4×10^{-8}.

On the same problem the measured accuracy–time frontier (Figure 2, left) has the lattice at 4.1×10^{-5} error in 0.03 seconds (at the twin-reference noise floor), direct MC at 1.2×10^{-4} in 15.7 seconds ($R = 10^7$), scrambled-Sobol direct QMC at 2.0×10^{-4} in 2.6 seconds (2^{20} points), GHK at 2.0×10^{-3} in 10.7 seconds ($R = 10^4$; over 8 seeds at $R = 10^3$ its error spans $[3.9\times10^{-3}, 1.3\times10^{-2}]$), and QMC-GHK (scrambled-Sobol uniforms in the same kernel) at 4.2×10^{-4} in 7.0 seconds ($R = 8192$), a five-to-sevenfold error improvement over plain GHK at equal R and still well off the lattice frontier. Below 10^{-3} the deterministic map separates decisively, consistent with $R^{-1/2}$ against a rapidly converging quadrature. One stronger stochastic baseline remains to be added: minimax tilting [9].

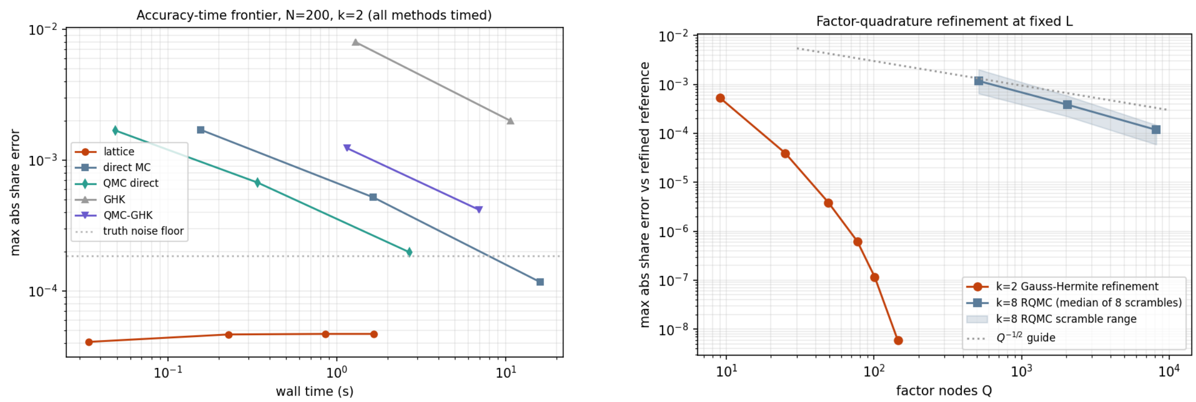

Figure 2: Left: measured accuracy–time frontier at $N = 200$, $k = 2$ (experiment 17); every method is timed on the same problem against twin 5×10^7-draw references, with direct simulation and QMC-GHK as first-class baselines. Right: factor-quadrature refinement at fixed L: Gauss–Hermite at $k = 2$, and at $k = 8$ the median and range over eight independent RQMC scrambles.

Derivatives. For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of μ whose derivatives are evaluated without resampling; analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. (Along a utility path at $N = 50$, the maximum second difference over dt^2 is 39.3 for fresh-draw GHK against 0.01 for both CRN-GHK and the lattice; the committed smoothness figure shows the curves.) These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws [3, 4]. The practical distinction is the character and controllability of the approximation error, quadrature resolution versus draw count, not the possibility of differentiation.

3 Calibration at $N = 1000$

At $N = 1000$, $k = 2$, with targets from 5×10^6 MC draws floored at 10^{-7} and renormalized: the inversion completes in 57 seconds. The iteration solves its own equations to 6.4×10^{-9} (a statement about the solver, not statistical accuracy, since the target carries sampling noise near 2×10^{-4}) and recovers the utilities of the well-resolved alternatives (target share above

3×10^{-4}) to maximum error 0.017. A replication study (experiment 16) repeats the inversion on a second $N = 1000$ problem against three independent 5×10^6-draw targets: 57 seconds each, 161–162 alternatives well-resolved (98.1% of share mass), recovery error maximum 0.015–0.023 and median ≈ 0.0027 across replicates, with the recovery-versus-share curve committed alongside.

Validation separates the error sources (experiment 21). A noiseless inverse test (targets generated by the lattice map itself, a deliberate inverse crime that isolates solver plumbing) recovers the utilities of every resolvable alternative to 6.5×10^{-7} at $N = 200$ and 1.2×10^{-7} at $N = 1000$ — solver error at the tolerance, so the ≈ 0.02 recovery in noisy tests is target sampling noise, not solver error. Alternatives with shares below the resolution floor are not chased, by design; their tier is reported separately. A robustness sweep of twenty problems varying utility spread, factor strength, D-heterogeneity, and rank converged in all twenty cases (6–31 iterations, recovery 0.004–0.024). Self-convergence certificates at large N bound the quadrature error the simulation references cannot resolve: refining to $(L = 3001, \mathrm{GH}21)$ moves the shares by 1.2×10^{-8} at $N = 1000$ and 5.7×10^{-8} at $N = 5000$.

A compiled implementation of the same pass (Rust, in the repository, with machine-precision parity and identical iteration counts) calibrates $N = 1000$ in 8 seconds and $N = 5000$ in 45 seconds; the single-threaded NumPy timings above remain the reported baseline.

Scaling is linear in N (experiment 19): calibration takes 48, 120, 330, and 639 seconds at $N = 1000$, 2000, 5000, 10000, a flat 11–15 forward-pass equivalents throughout, with recovery quality unchanged (median error near 0.003, 96–99% of share mass identified). At $N = 5000$ a single matched-accuracy forward evaluation by the per-alternative simulator extrapolates to roughly fifty hours and calibration needs about fifteen of them, so the same calibration through simulation extrapolates to weeks. Numerical well-posedness matches Theorem 1: the reduced Jacobian assembled from N JVP calls at $N = 50$ is symmetric to 1.7×10^{-18}, has row-sum defect 6.2×10^{-17}, and reduced eigenvalues in $[2.5 \times 10^{-5}, 0.22]$, positive definite as proved. The gap to simulation here is structural, not incremental, and simple arithmetic shows why (labeled as arithmetic, not measurement). An iterative inversion evaluates the forward map $\sim$15 times and cannot drive its residual below the per-evaluation noise; simulation noise on shares scales as $\sqrt{p_{\max}/R}$, so a share-residual tolerance τ costs $R \approx p_{\max}/\tau^2$ draws per evaluation. Under independent direct simulation at the measured $N = 5000$ throughput, this variance calculation prices one forward evaluation at this solver's model-side consistency ($\tau \sim 10^{-6}$) in weeks and the calibration in months to a year; variance reduction and common random numbers improve the constants but not the τ^{-2} scaling; even $\tau = 10^{-5}$ prices the calibration in days against our 330 seconds; and at the crude $\tau = 10^{-4}$ where simulation-in-the-loop first becomes feasible, the calibrated model mismatches its own targets at 10^{-4} and the derivative noise forces stochastic-approximation methods. The quadrature route's cost is independent of τ until resolution binds. We are unaware of a reported probit share inversion at this scale in simulation-based practice, and this scaling is presumably why.

Removal ensemble (by-product). The cached field also yields all N single-removal share vectors in $O(QN^2L)$. Against the strongest simulation comparator, which reuses each draw's winner and runner-up to answer every removal at once in $O(RN + N^2)$, the deterministic ensemble is about 3× more accurate at matched wall time at $N = 200$ (4.2×10^{-5} versus 1.2×10^{-4}; experiment 18). This is a side capability, not the paper's case.

4 Boundary experiments

General covariance through the factor approximation. The method requires $\Sigma \approx VV^\top + \mathrm{diag}(D)$; GHK does not. Choices depend on (μ, Σ) only through $P\mu$ and $P\Sigma P$, $P = I - \mathbf{1}\mathbf{1}^\top/N$, because the argmax is determined by pairwise differences: a common factor loading $V \to V + \mathbf{1}c^\top$ shifts every conditional location equally and cannot move the argmax (a machine-precision identity), so the factor fit must be run in this choice-relevant quotient space. Fitting raw Σ can spend scarce rank on an irrelevant common component ($\Sigma = \tau^2\mathbf{1}\mathbf{1}^\top + \beta\beta^\top + D$ with large τ: the raw rank-1 fit takes $\mathbf{1}\mathbf{1}^\top$ and misses $\beta\beta^\top$ entirely; on such an example the contrast-space fit cuts rank-1 share error from 4.6×10^{-2} to 8.3×10^{-3}). The asymmetry matters: only the common *factor* direction is irrelevant; a common addition to D inflates every difference variance and is choice-relevant, so D must come from the quotient fit. The fit itself is an iterative principal-factor heuristic applied to $P\Sigma P$ (eigendecompose $P\Sigma P - \mathrm{diag}(D)$, take the top k directions, refit D from the diagonal with a 10^{-3} floor, iterate to 10^{-10}), followed by centering and SVD canonicalization of the loadings (ordered columns, fixed signs); it is not certified to minimize a projected norm, and the covariance-quotient residual $\|P(\Sigma - \hat{\Sigma})P\|_F/\|P\Sigma P\|_F$ is reported alongside every share error in the committed output.

With this fit: correlation matrices at spectral decay $\lambda_m \propto m^{-\gamma}$, $\gamma \in \{0.5, 1.5, 3\}$, at $N = 50$, sharing one orthogonal eigenbasis across γ so decay comparisons are not confounded with basis realization. The power law holds before diagonal standardization, which perturbs the spectrum; the actual leading eigenvalues (3.7, 17.1, 34.2 at $\gamma = 0.5, 1.5, 3$) are printed by the script. Share error is measured against an 8×10^6-draw reference, with GHK at the exact Σ for comparison; the plotted quantity is the rank-k approximation error of the stated procedure, not a minimum over factor models. Rank-8 errors: 1.8×10^{-3}, 3.7×10^{-3}, and 3.1×10^{-3} at $\gamma = 0.5, 1.5, 3$ from $k = 1$ starting points of 2.3×10^{-3}, 1.5×10^{-2}, 7.1×10^{-2}.

Decomposing the error with a second simulation under the fitted factor model: integration error is flat at 2–7×10^{-4} across every (γ, k) tested, so the rank-k error is essentially covariance-fit error: rank, not quadrature, is the binding constraint. Against GHK at $R = 10^4$ (3.2, 2.3, 4.3×10^{-3}), rank-8 wins at $\gamma = 0.5$ and 3 but *loses at intermediate decay* $\gamma = 1.5$ (3.7×10^{-3} versus 2.3×10^{-3}): for this construction the intermediate case is empirically hardest. (Diagonal standardization perturbs the nominal power law and the leading eigenvalue varies from 3.7 to 34.2 across the three cases, so this is not a clean one-parameter family. Replication over three independent eigenbases: rank-8 beats GHK at $R = 10^4$ in 8 of 9 basis–decay cases, with the single loss at intermediate decay, so the intermediate construction is empirically hardest but not a stable GHK-wins regime.) Nor is this wall-time-matched: the $k = 8$ RQMC evaluation takes 15–24 s against 0.5–0.9 s for GHK at $R = 10^4$, so at $N = 50$ GHK is the cheaper route to 3–4×10^{-3}; the lattice's advantage is large N at small k. Where accuracy demands exceed the achievable rank-k error, GHK's arbitrary-Σ generality is the right tool.

Substitution: a 2×2 factorial illustration. The design varies factor structure and noise family one axis at a time, holding the idiosyncratic scale common. Coupling either axis with alternative-specific scales would confound the comparison: an alternative-specific-scale Gumbel race is not mixed logit, and heteroskedasticity alone already breaks IIA. A separate heteroskedasticity axis is left to future replication.

Ground truth ($N = 50$) is misspecified for every candidate: $t(5)$ factors (unit variance) and skew-normal idiosyncratic noise (shape $a = 3$), standardized to mean zero and unit variance, homoskedastic, with $\mu^\star \sim N(0, 1)$ and oracle loadings $v^\star_{ir} \sim N(0, 0.36/k)$, $k = 2$. (The race is computed min-wins, so positive skew in race coordinates is *left*-skew in utility coordinates.) The four candidates hold the idiosyncratic scale fixed and vary $\{\text{Gumbel}, \text{Gaussian}\}$ base $\times$ $\{V = 0, V = V^\star\}$; the Gumbel base is variance-standardized to one, so the family axis does not alter the factor-to-idiosyncratic variance ratio, and the candidate factor law is Gaussian in all cases (the truth's $t(5)$ factor law is misspecified for every candidate). This is an *oracle-loading* design: both factor candidates receive the true $V^\star$ rather than estimating it. All four are calibrated to the same full-menu shares (residuals $\leq 4 \times 10^{-10}$).

Deletion truths come from 2×10^7 *common* utility draws with the top three alternatives retained per draw, from which every single-deletion and pair-deletion truth follows with common random numbers. Of 150 deletion blocks (50 singles, 100 fixed-seed random pairs), 45 with deleted mass below 0.05% are excluded as unresolvable; of the 105 scored blocks, 83 fall in the three reported strata (26/29/28 at >10% / 2–10% / 0.5–2%) and the remaining 22 have deleted mass between 0.05% and 0.5%. The score per deletion block B is the misallocated fraction of the redistributed mass,

$$\mathrm{TV}\Big(q^{(-B)}_{\text{model}}, q^{(-B)}_{\text{truth}}\Big) \Big/ \textstyle\sum_{i\in B} p_i,$$

averaged within strata:

	mass > 10%	2–10%	0.5–2%
independent Luce (= plain logit)	0.229	0.245	0.261
independent probit	0.210	0.232	0.230
factor mixed logit	0.114	0.146	0.199
factor probit	**0.045**	**0.062**	**0.094**

The factorial decomposition (top two strata): the factor increment is dominant in both families (+0.115/+0.099 within Gumbel, +0.165/+0.170 within Gaussian); the family increment alone is small ($+0.019/+0.013$ at $V = 0$) but grows with factors present ($+0.069/+0.084$), a negative interaction ($-0.050/-0.071$): factor structure and the Gaussian base are complements in this design. Caveats: a single truth design and seed, and a skew-normal truth that is Gaussian-adjacent. The figure shows every scored single deletion individually (27 of the 50; the rest fall below the 0.05% floor); singles and pairs score similarly (factor probit 0.080 versus 0.076 mean misallocated fraction) and are summarized separately in the committed output.

5 Related work

GHK and simulated likelihood are standard [2, 3, 4]. Bhat's MACML [7] and the approximations surveyed by Connors, Hess and Daly [8] trade simulation for analytic marginals, still one alternative at a time. Botev [9] sharpened Gaussian-restriction simulation with minimax tilting.

Factor dimension reduction for probit is old. Butler and Moffitt [5] integrated one factor by quadrature; Elrod and Keane [6] estimated factor-analytic probit for panel data; Loaiza-Maya and Nibbering [18] pursue large choice sets by Bayesian estimation. The lattice survival-product identity and its inverse transform come from racing [10].

Inversion theory is likewise established. Norets and Takahashi [17] prove surjectivity of the utility-to-probability map; Chiong, Galichon and Shum [16] invert through the convex conjugate; Berry, Levinsohn and Pakes [11] built demand estimation on the logit-side inversion, and

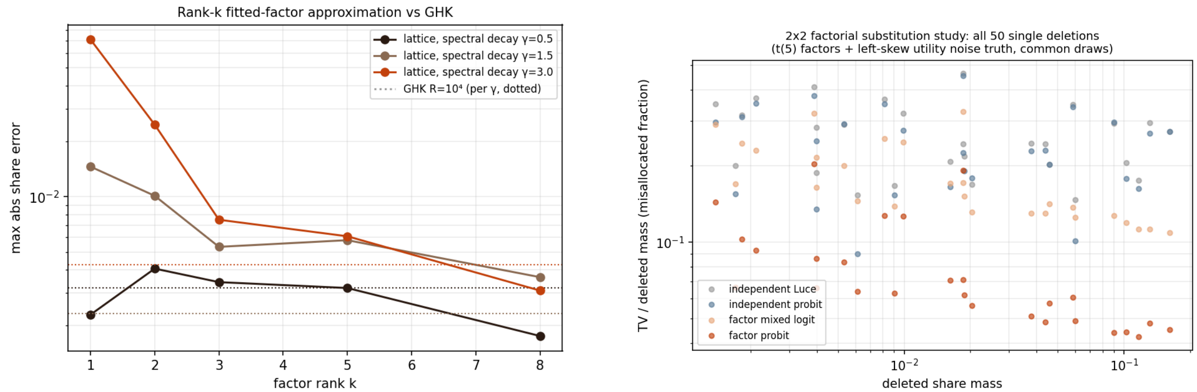

Figure 3: Left: rank-k contrast-space approximation error under spectral decay γ (dotted: GHK at $R = 10^4$, per γ; not wall-time matched). Right: the 2×2 factorial substitution study, with every scored single-deletion block shown individually (27 of 50).

Conlon and Gortmaker [12] maintain its modern practice. Chia [13] differentiates logit BLP automatically; Huch and Keane [14] amortize correlated-choice probabilities with neural networks, evidence of current demand for this computation.

The novelty here is accordingly none of: dimension reduction, the order-statistic integral, or well-posedness of inversion. It is the all-alternatives assembly in $O(QNL)$, its large-N implementation, the $O(QNL)$ Jacobian–vector product, and the shared-field reuse for calibration and counterfactuals.

6 Discussion

Estimation is the natural next step. With the reduced coordinates of Proposition 3, gradients of an outer likelihood or GMM objective pass through the calibration fixed point by implicit differentiation, a route to differentiable probit demand estimation at large N. Estimating (V, D) rather than supplying them raises the usual identification cautions: scale normalization, factor rotation, and the fact that only difference covariances are identified.

The pass itself can be accelerated further: its two stages are smooth-kernel sums whose matrices are numerically low-rank, and a Chebyshev-separated evaluation (in the spirit of black-box fast multipole methods) reduces each node's cost from $O(NL)$ to $O(r(N+L))$; a committed prototype (experiment 20) reproduces the exact pass to 6×10^{-5} at $45\times$ speed for $N = 5000$, with exponential convergence in r. Development to full generality is left to a companion note.

Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities. A common-scale Gumbel base with zero loadings reproduces the Luce model exactly, yielding correlated-softmax generalizations.

Neither method class dominates the other. Simulation owns arbitrary Σ and coarse accuracy; the shared field owns the factor form, tight accuracy, derivatives, and calibration; the boundary experiments draw the border at intermediate spectral decay. We leave to future work the minimax-tilting baseline, replication of the substitution factorial across truth designs, and a walk-forward calibration on real assortment data.

Reproducibility

All numbers come from committed, seeded scripts with correctness anchors run before any comparison (repository experiments 13–21; a single manifest, `experiments/run_all_paper.py`, regenerates every table and figure; the exact commit is recorded in the repository README and an immutable tag is pinned at circulation). The algorithm ships in the `thurstone` package, and package–benchmark agreement is itself a reported number. Hardware and software: Apple M4, 16 GB, Python 3.12.9, NumPy 2.4.6, SciPy 1.18.0; experiment 17 enforces single-threaded BLAS via `threadpoolctl`; sub-second timings are warmed medians of three runs, and long simulations are single realizations, disclosed as such. Monte Carlo chunk sizes derive from an explicit 1.5 GB memory budget, and peak resident memory stays below 2 GB.

References

[1] L. L. Thurstone. A law of comparative judgment. *Psychological Review* 34 (1927) 273–286.

[2] J. Geweke, M. Keane, D. Runkle. Alternative computational approaches to inference in the multinomial probit model. *Review of Economics and Statistics* 76 (1994) 609–632.

[3] V. Hajivassiliou, D. McFadden. The method of simulated scores for the estimation of LDV models. *Econometrica* 66 (1998) 863–896.

[4] K. Train. *Discrete Choice Methods with Simulation*, 2nd ed. Cambridge, 2009.

[5] J. S. Butler, R. Moffitt. A computationally efficient quadrature procedure for the one-factor multinomial probit model. *Econometrica* 50 (1982) 761–764.

[6] T. Elrod, M. P. Keane. A factor-analytic probit model for representing the market structure in panel data. *Journal of Marketing Research* 32 (1995) 1–16.

[7] C. Bhat. The maximum approximate composite marginal likelihood (MACML) estimation of multinomial probit-based unordered response choice models. *Transportation Research B* 45 (2011) 923–939.

[8] R. Connors, S. Hess, A. Daly. Analytic approximations for computing probit choice probabilities. *Transportmetrica A* 10 (2014) 119–139.

[9] Z. Botev. The normal law under linear restrictions: simulation and estimation via minimax tilting. *JRSS-B* 79 (2017) 125–148.

[10] P. Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM J. Financial Mathematics* 12 (2021) 295–317.

[11] S. Berry, J. Levinsohn, A. Pakes. Automobile prices in market equilibrium. *Econometrica* 63 (1995) 841–890.

[12] C. Conlon, J. Gortmaker. Best practices for differentiated products demand estimation with pyblp. *RAND Journal of Economics* 51 (2020) 1108–1161.

[13] A. Chia. Automatically differentiable random coefficient logistic demand estimation. arXiv:2106.04636 (2021).

[14] E. Huch, M. Keane. Amortized inference for correlated discrete choice models via equivariant neural networks. arXiv:2603.24705 (2026).

[15] D. McFadden. Econometric models of probabilistic choice. In C. Manski, D. McFadden (eds.), *Structural Analysis of Discrete Data*, MIT Press, 1981.

[16] K. Chiong, A. Galichon, M. Shum. Duality in dynamic discrete-choice models. *Quantitative Economics* 7 (2016) 83–115.

[17] A. Norets, S. Takahashi. On the surjectivity of the mapping between utilities and choice probabilities. *Quantitative Economics* 4 (2013) 149–155.

[18] R. Loaiza-Maya, D. Nibbering. Scalable Bayesian estimation in the multinomial probit model. *Journal of Business & Economic Statistics* 40 (2022) 1678–1690.

[19] The `allocation` package: Thurstone-tilt portfolio construction. `https://github.com/microprediction/allocation`.